

BIWEEKLY ENGINEERING REPORT · SEPTEMBER 9, 2026

Multimodal Document Understanding Harness

Baseline vs. GLM-OCR — Benchmarked Across TAT-DQA & UniDoc-Bench



Ingest



Extract



Retrieve



Read



Evaluate

Environment: Python 3.12 · Gemini Vision (gemini-3.5-flash-lite / gemini-3.6-flash) · RRF Hybrid Retrieval

What This Report Covers



01

Where We Started

Six failure modes suppressing baseline & GLM-OCR performance



02

Ten Engineering Fixes

Point-wise interventions across ingestion, retrieval, and reading



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Metric Definitions

How retrieval, QA, and LLM-judge scores are computed



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Four Benchmark Runs

Baseline & GLM-OCR, each on TAT-DQA and UniDoc-Bench



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Cross-Pipeline Synthesis

A single comparison matrix across all four runs



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Key Takeaways

What changed, what's next

Where We Started: Six Failure Modes



GLM-OCR Local Crash

Missing torchvision/weights silently degraded every page to bare PyMuPDF text — no tables, no figures.



GLM-OCR Visual Blindness

algorithms/image_extraction/glm_ocr never existed, so the OCR backend had no image_extractor at all.



Baseline Vector Blindness

page.get_images() returned zero on vector-drawn charts and logos, skipping figure captioning entirely.



Reader Context Starvation

A 4,000-character cap truncated multi-chunk figure captions before they reached the LLM reader.



Rate Limit Throttling

gemini-2.5-flash's 20 req/day free-tier quota triggered repeated 429 Resource Exhausted errors.



TAT-DQA 100% Smoke-Test Mirage

A single-document candidate pool (Total chunks: 1) made Hit@1 = 100% mathematically guaranteed.

How Bad It Actually Was

6.1%

Initial UniDoc visual-query score

Queries 1–5 in finance.json are 100% image-only visual queries — yet the legacy runner scored just 6.1% in aggregate.

100%

TAT-DQA smoke-test mirage

--limit 5 indexed a single document, so the retrieval pool held only that one doc — Hit@1 was guaranteed, not earned.

ROOT CAUSES BEHIND THE 6.1% SCORE

Vector Graphics Blindness

Charts and logos in modern PDFs are drawn with vector commands, not raster Image XObjects — `page.get_images()` saw nothing.

Context Starvation

Figure captions landing in chunk 6 or 7 were cut off by the reader's 4,000-character truncation limit.

Imprecise Prompting

The generic “Describe this figure” prompt produced prose that omitted exact numbers and axis labels.

Stale Negative Cache

Cached NOT_FOUND answers kept returning even after the missing context was finally supplied.

Ten Engineering Fixes

1 Vector Drawing Density Scanner

3 Reader Context Expansion (4k → 36k)

5 GLMOCRFigureExtractor Creation

7 Full-Page Multimodal Prompt Upgrade

9 Gemini Model Prioritization

2 Verbatim Multimodal Prompting

4 Negative Cache Sanitation

6 OCR Stage Registration

8 Bilateral Runner Wiring

10 Multi-Doc Sampling CLI Flags

The Four Fixes That Mattered Most



Vector Drawing Density Scanner

Pages with ≥ 5 vector drawings or chart keywords are rendered at 150 DPI and routed to the visual figure extractor — catching charts with zero raster objects.



Verbatim Multimodal Prompting

Replaced “Describe this figure” with a transcription prompt forcing exact labels, axes, and numbers instead of subjective prose.



Reader Context Expansion

Truncation threshold raised from 4,000 to 36,000 characters, so tabular context and visual captions reach the LLM reader together.



Gemini Model Prioritization

Fallback order set to gemini-3.5-flash-lite → gemini-flash-lite-latest → gemini-3.6-flash, eliminating all 429 quota errors.

How We Measure Success



Retrieval Layer

Hit@1

Top-ranked chunk matches the gold document

Hit@5 / Hit@10

Gold document appears in top 5 / top 10

MRR

Average of $1 / \text{rank of the first relevant hit}$

NDCG@10

Ranking quality, discounted by position



True E2E QA (TAT-DQA)

Exact Match

Normalized string equality vs. gold

Numeric EM

Equality after stripping \$, commas, units

Set EM

All expected spans present, any order

Token F1 / Semantic

Bag-of-word overlap / embedding similarity

Headline Score

$0.30 \cdot \text{Retrieval} + 0.35 \cdot \text{F1} + 0.35 \cdot \text{Numeric EM}$



LLM-as-a-Judge (UniDoc)

Factual Accuracy

Correctness vs. gold facts in the source

Completeness

All requested entities/dates addressed

Answer Relevance

Directly answers, no digression

Overall Judge Score

Holistic blend of the three dimensions

Reformed Baseline on TAT-DQA



RETRIEVAL LAYER	
Hit@1	80.0%
Hit@5	100.0%
Hit@10	100.0%
MRR	0.9000
NDCG@10	0.9262
QA LAYER (TRUE END-TO-END)	
Exact Match (EM)	80.0%
Numeric EM	80.0%
Set EM (Multi-Span)	80.0%
Token F1	81.0%
Precision / Recall	80.6% / 82.9%
Semantic Score	80.0%
Headline Score	80.1%

80.1%

Headline Score — a reformed baseline the harness can trust

KEY ENGINEERING NOTE

Table Header Injection (THI) converts TableFormer grid rows into natural-language chunks that retain column headers and fiscal-period context — the single largest contributor to the 81.0% Token F1.

Reformed Baseline on UniDoc-Bench



RETRIEVAL LAYER	
Hit@1 / Hit@5 / Hit@10	100.0%
MRR / NDCG@10	1.0000
GEMINI LLM JUDGE	
Overall Judge Score	76.0%
Factual Accuracy	74.0%
Completeness / Coverage	76.0%
Answer Relevance	80.0%
REAL PDF SCORE	
Average on 4 Real PDFs	95.0%

PER-QUERY EVIDENCE

Q1 · Peru Real GDP	90%
Q2 · RIT Capital	0%
Q3 · Research Affiliates	100%
Q4 · GWAC Home Loans	90%
Q5 · Keefe McCullough	100%

Q2's miss traces to a dataset issue: the backing PDF is mislabeled, not a pipeline failure.

GLM-OCR on TAT-DQA



Metric	5-Query Smoke Test	30-Query Multi-Doc
Hit@1	100.0%	90.0%
Hit@5	100.0%	100.0%
Hit@10	100.0%	100.0%
MRR	1.0000	0.9317
NDCG@10	1.0000	0.9483
Numeric EM	100.0%	76.7%
Exact Match	100.0%	60.0%
Set EM	100.0%	46.7%
Token F1	100.0%	68.1%
Semantic Score	100.0%	81.7%
Headline Score	100.0%	77.6%



Why the 5-query column reads 100.0% across the board: --limit 5 indexed a single document, guaranteeing Hit@1 — the 30-query multi-document run at right is the real signal.

GLM-OCR on UniDoc-Bench



RETRIEVAL LAYER	
Hit@1 / Hit@5 / Hit@10	100.0%
MRR / NDCG@10	1.0000
GEMINI LLM JUDGE	
Overall Judge Score	76.0%
Factual Accuracy	76.0%
Completeness / Coverage	77.0%
Answer Relevance	76.0%
REAL PDF SCORE	
Average on 4 Real PDFs	95.0%

PER-QUERY EVIDENCE	
Q1 · Peru Real GDP	85%
Q2 · RIT Capital	0%
Q3 · Research Affiliates	100%
Q4 · GWAC Home Loans	100%
Q5 · Keefe McCullough	95%

Q2 again fails on a mislabeled backing PDF, not a retrieval or reading defect.

Cross-Pipeline Comparison Matrix

Metric	Baseline TAT-DQA	GLM-OCR TAT-DQA	Baseline UniDoc	GLM-OCR UniDoc
RETRIEVAL LAYER				
Hit@1	80.0%	90.0%	100.0%	100.0%
Hit@5	100.0%	100.0%	100.0%	100.0%
MRR	0.9000	0.9317	1.0000	1.0000
NDCG@10	0.9262	0.9483	1.0000	1.0000
QUANTITATIVE QA				
Numeric EM	80.0%	76.7%	N/A	N/A
Exact Match	80.0%	60.0%	N/A	N/A
Token F1	81.0%	68.1%	22.5%*	19.2%*
Headline Score	80.1%	77.6%	44.2%*	44.4%*
LLM-AS-A-JUDGE				
Overall Judge	N/A	N/A	76.0%	76.0%
Factual Accuracy	N/A	N/A	74.0%	76.0%
Completeness	N/A	N/A	76.0%	77.0%
Backing PDF Score	N/A	N/A	95.0%	95.0%

*UniDoc Token F1 / Headline Score are compressed by prose-length disparity vs. single-span answers — the Gemini LLM Judge (76.0%) is the reliable read on UniDoc quality.

Key Engineering Takeaways



Multimodal Parity Achieved

GLM-OCR now matches the reformed baseline's visual understanding — 95.0% accuracy transcribing vector charts, line graphs, and logos on backing PDFs.



Zero Quota Exhaustion

Reprioritizing to gemini-3.5-flash-lite and gemini-flash-lite-latest eliminated every 429 Resource Exhausted error across both benchmarks.



Multi-Document Robustness

The expanded 30-query, multi-document TAT-DQA test held 90.0% Top-1 accuracy — confirming RRF hybrid retrieval scales past single-doc smoke tests.

Next: extend the 30-query multi-document protocol to UniDoc-Bench, and resolve the Q2 dataset mislabeling.